Modeling spatial data February 20, 2026

1 Assignment description

- Submission deadline: *March 2, 2026, 12:00.*
- Grades released: *March 9, 2026, 12:00.*
- The assignment consists of two parts: main questions and resit questions. For the first submission, only the main questions must be completed.

  Failure to submit solutions to the main questions by the deadline results in loss of eligibility for the resit.
- If the main part is failed, one resit opportunity is available by submitting solutions to the resit questions. The resit deadline is *March 18, 2026, 11:00.*
- *Strict* format requirements:
  (a) Minimum font size: 11pt.
  (b) There is no strict page limit; however, explanations should be clear and concise.
  (c) Submit via Canvas Assignments and include:
    - a written report only in PDF format;
    - a zip file containing the code (in any language) required to replicate the results.

2 Main questions

This section contains three main questions for the first sit. All related articles are available on the Canvas course page.

Notation: All notation, unless otherwise stated, follows the lecture. In particular, let T denote the sample size throughout.

Question 1 (40 points)

In this exercise, we focus on static SAR models. Load the datasets "iowa_yield_05_10.csv" and "iowa_spatial_weight_row_normalized.csv". The file "iowa_yield_05_10.csv" contains three columns: the first lists the names of counties in Iowa, the second reports crop yields (bushels per acre) in 2005, and the third reports the observations in 2010. The file "iowa_spatial_weight_row_normalized.csv" contains the spatial weight matrix constructed using inverse distance and subsequent row normalization. Specifically, we compute the distance between each pair of counties, take its inverse to form the inverse-distance matrix, and then row-normalize the resulting matrix.

(a) First, use the 2010 observations to replicate the results presented on page 29 of the lecture slides on modeling spatial data. Specifically, implement the LR test for spatial autoregression and estimate the static SAR model without regressors.

Then use the 2005 observations to repeat the estimation and testing procedures. What do you conclude for 2005?

(b) We now further investigate the properties of the spatial weight matrix through simulation. Consider a simple experiment. Take the SAR model without regressors and generate the data $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)^\top$ as follows:

$$\boldsymbol{y} = \rho \boldsymbol{W} \boldsymbol{y} + \boldsymbol{e}, \qquad \boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n),$$

with the true parameters $\rho = 0.5$, $\sigma^2 = 1$, $n \in \{1000, 2000\}$. The spatial weight matrix $\boldsymbol{W}$ is generated according to the following four steps: (1) generate an $n \times n$ matrix with i.i.d. entries from $\mathrm{U}[0, 1]$; (2) take the absolute value of the resulting matrix to ensure all elements are positive; (3) set the diagonal elements to zero (i.e., no self-spillovers); and (4) normalize the matrix either by row normalization or by dividing it by its spectral radius. Note that once $\boldsymbol{W}$ and $\boldsymbol{e}$ are generated, $\boldsymbol{y}$ can be obtained as $\boldsymbol{y} = (\boldsymbol{I}_n - \rho \boldsymbol{W})^{-1} \boldsymbol{e}$.

Now use the generated data $\boldsymbol{y}$ to estimate the parameters (ρ, σ^2) by fitting the SAR model. Are the resulting estimates accurate?

Since a single simulation provides limited information about finite sample performance, conduct a Monte Carlo study. Depending on your computational resources, you may run, for example, 200 or 1000 replications. Plot the histograms of the resulting estimates, normalized to represent probability density functions. What do you observe? Are the histograms concentrated around the true parameter values? If not, does increasing n from 1000 to 2000 improve the accuracy, or does it result in worse or unchanged performance?

(c) Can you explain your simulation results by relating them to the assumptions in Lee (2004)?

(d) In particular, can you also provide a small-scale simulation study to support your explanation in Part (c)?

Question 2 (60 points)

In this exercise, we focus on dynamic spatial models with homogeneous spatial spillover effects. Load the datasets "cds_data.xlsx" and "cds_spatialweights.xlsx". These two datasets are the same as those used in Blasques et al. (2016); see also pages 54–55 of the lecture slides for explanations. The dataset "cds_data.xlsx" contains the CDS data (columns 1–8), i.e., our $\boldsymbol{y}_t$, as well as regressors $\boldsymbol{X}_t$ including three financial market variables that are common across countries (columns 9–11) and country-specific regressors: the main stock indices (columns 12–19) and interest rate spreads (columns 20–27). The dataset "cds_spatialweights.xlsx" contains the observed spatial weight matrix measuring economic distance. Note that this observed weight matrix has not yet been normalized.

(a) Use the provided data to replicate the figure based on Gaussian score-driven dynamics on page 56 of the lecture slides. Note that the parameter spaces and link function are given on page 55, and the detailed construction is provided on page 52.

Moreover, try using different specification of parameter spaces and explain whether you observed substantial changes.

Finally, implement the QLR test for time variation. The simulated critical values can be found in Table 1 below. Importantly, explain which specific critical values do you use. This depends on the specification of your parameter spaces $\mathcal{B}$ and Θ_α.

(b) Provide detailed steps to derive the multivariate t score-driven dynamics on pages 40–41. Importantly, if you use specific formulas, state explicitly which formulas you apply.

(c) Use the Gaussian and t score-driven models to replicate Figure 4 in Blasques et al. (2016). Note that the scale matrix $\boldsymbol{\Sigma}$ in the multivariate t distribution is restricted to $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}_n$, as in the Gaussian dynamics. If you are not able to obtain exactly the same results, aim to match them as closely as possible. State explicitly all details required to reproduce your results without referring to your code. In particular, your report should include the full specification of parameter spaces, link functions, initial values, and any other choices needed for complete replication.

(d) Implement the QLR test for time variation as in Part (a), but now using multivariate t score-driven dynamics (with $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}_n$) instead of Gaussian dynamics. Provide the detailed steps leading to your test; you may follow the structure of the example on pages 52–53 of the lecture slides. What is the value of your test statistic? Which critical values do you use, and why do you choose them? What do you conclude?

(e) On page 52 of the lecture slides, it is stated that "Even when the true parameter is time-constant, naively applying score-driven models may produce pronounced time-varying patterns." We now verify this statement. To this end, consider the following simple simulation experiment. Generate observations $\{y_1, \ldots, y_T\}$ with $T = 500$, where

$$y_t = 0.2 + u_t, \qquad u_t \overset{\text{i.i.d.}}{\sim} t_\nu, \qquad \nu = 10.$$

Clearly, the location is time-constant. Now naively apply the *univariate* t-location model you coded for Question 3 in Assignment 1 with $\mathcal{B} \subset [0, 1]$ and $\Theta_\alpha = [0, 2]$. Report all other modeling and implementation choices, including any additional parameter spaces. Run the estimation multiple times and plot the estimated location path $\hat{f}_t$. Do you agree with this statement? If so, can you provide a plausible explanation for why this can occur?

Enjoy and Success!

3 Resit questions

Question R1

Load "iowa_precipitation_05_10.csv" in addition to the datasets "iowa_yield_05_10.csv" and "iowa_spatial_weight_row_normalized.csv".

R1(a) Repeat Part (a) in Question 1 of Section 2 by including precipitation as a regressor. Specifically, implement the LR test for spatial autoregression and estimate the static SAR model with precipitation included as an explanatory variable.

R1(b) Construct a 95%-level confidence interval for ρ using the results on pages 22–23 of the lecture slides.

Question R2

Repeat Parts (a), (c), and (d) in Question 2 of Section 2, but use a full matrix $\boldsymbol{\Sigma}$ instead of the restricted specification $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}_n$.

References

Blasques, F., S. J. Koopman, A. Lucas, and J. Schaumburg (2016). Spillover dynamics for systemic risk measurement using spatial financial time series models. *Journal of Econometrics 195*(2), 211–223.

Lee, L.-F. (2004). Asymptotic distributions of quasi-maximum likelihood estimators for spatial autoregressive models. *Econometrica 72*(6), 1899–1925.

Table 1: Simulated critical values for the limiting distribution of the QLR test for absence of score-driven dynamics with $\Theta_\alpha = [\alpha_L, \alpha_U]$ for $\alpha_U > 0$ and $\mathcal{B} = [\beta_L, \beta_U]$.

		$\alpha_L = 0$			$\alpha_L < 0$					$\alpha_L = 0$			$\alpha_L < 0$		
β_L	β_U	10%	5%	1%	10%	5%	1%	β_L	β_U	10%	5%	1%	10%	5%	1%
0	0.995	3.365	4.719	7.855	4.437	5.850	9.127	-0.995	0.995	3.913	5.362	8.714	5.014	6.507	9.909
0	0.990	3.266	4.613	7.696	4.416	5.816	9.048	-0.990	0.990	3.842	5.278	8.629	4.996	6.481	9.886
0	0.980	3.166	4.502	7.540	4.371	5.753	8.962	-0.980	0.980	3.760	5.191	8.506	4.964	6.442	9.814
0	0.970	3.092	4.417	7.425	4.322	5.691	8.854	-0.970	0.970	3.703	5.128	8.423	4.936	6.405	9.746
0	0.960	3.037	4.355	7.363	4.278	5.638	8.794	-0.960	0.960	3.657	5.076	8.343	4.905	6.376	9.706
0	0.950	2.989	4.308	7.277	4.247	5.587	8.731	-0.950	0.950	3.614	5.021	8.298	4.873	6.336	9.673
0	0.900	2.810	4.106	7.080	4.084	5.403	8.502	-0.900	0.900	3.456	4.840	8.116	4.748	6.155	9.470
0	0.850	2.675	3.949	6.930	3.958	5.236	8.300	-0.850	0.850	3.319	4.688	7.909	4.626	6.000	9.267
0	0.800	2.568	3.831	6.813	3.840	5.114	8.169	-0.800	0.800	3.197	4.545	7.699	4.499	5.857	9.100
0	0.750	2.490	3.732	6.720	3.737	5.007	8.043	-0.750	0.750	3.087	4.411	7.546	4.382	5.728	8.919
0	0.700	2.417	3.644	6.619	3.651	4.908	7.955	-0.700	0.700	2.981	4.300	7.378	4.271	5.614	8.767
0	0.650	2.341	3.567	6.530	3.568	4.805	7.862	-0.650	0.650	2.880	4.183	7.265	4.165	5.483	8.598
0	0.600	2.272	3.488	6.451	3.489	4.734	7.730	-0.600	0.600	2.778	4.058	7.133	4.052	5.367	8.427
0	0.550	2.218	3.413	6.385	3.411	4.656	7.640	-0.550	0.550	2.689	3.944	6.981	3.954	5.226	8.296
0	0.500	2.158	3.339	6.296	3.342	4.579	7.529	-0.500	0.500	2.592	3.829	6.853	3.841	5.109	8.145
0	0.450	2.099	3.264	6.204	3.273	4.499	7.425	-0.450	0.450	2.499	3.731	6.722	3.736	4.987	8.009
0	0.400	2.042	3.207	6.102	3.210	4.419	7.335	-0.400	0.400	2.402	3.631	6.632	3.632	4.872	7.887
0	0.350	1.988	3.153	6.028	3.146	4.359	7.245	-0.350	0.350	2.307	3.524	6.529	3.522	4.763	7.745
0	0.300	1.932	3.094	5.978	3.090	4.275	7.166	-0.300	0.300	2.214	3.410	6.349	3.407	4.645	7.585
0	0.250	1.884	3.025	5.898	3.024	4.205	7.095	-0.250	0.250	2.120	3.297	6.232	3.301	4.521	7.432
0	0.200	1.835	2.969	5.811	2.968	4.135	6.999	-0.200	0.200	2.026	3.188	6.090	3.189	4.400	7.295
0.1	0.995	3.334	4.679	7.784	4.421	5.821	9.086	-0.1	0.995	3.403	4.757	7.896	4.463	5.868	9.185
0.1	0.990	3.226	4.572	7.619	4.395	5.788	8.997	-0.1	0.990	3.308	4.655	7.751	4.443	5.845	9.100
0.1	0.980	3.122	4.452	7.457	4.338	5.710	8.893	-0.1	0.980	3.206	4.537	7.609	4.404	5.792	8.997
0.1	0.970	3.044	4.367	7.377	4.288	5.652	8.789	-0.1	0.970	3.133	4.467	7.507	4.363	5.735	8.914
0.1	0.960	2.983	4.304	7.290	4.243	5.598	8.740	-0.1	0.960	3.086	4.413	7.429	4.331	5.687	8.844
0.1	0.950	2.937	4.245	7.222	4.204	5.541	8.670	-0.1	0.950	3.037	4.366	7.380	4.300	5.644	8.799
0.1	0.900	2.749	4.041	6.983	4.030	5.329	8.423	-0.1	0.900	2.867	4.173	7.176	4.146	5.469	8.604
0.1	0.850	2.614	3.877	6.842	3.887	5.154	8.230	-0.1	0.850	2.743	4.032	7.045	4.029	5.316	8.424
0.1	0.800	2.504	3.757	6.707	3.759	5.025	8.063	-0.1	0.800	2.643	3.913	6.900	3.920	5.193	8.287
0.3	0.995	3.265	4.594	7.702	4.370	5.770	8.944	-0.3	0.995	3.475	4.840	8.018	4.528	5.945	9.264
0.3	0.990	3.156	4.483	7.510	4.333	5.713	8.861	-0.3	0.990	3.389	4.754	7.855	4.513	5.926	9.231
0.3	0.980	3.034	4.348	7.331	4.264	5.619	8.746	-0.3	0.980	3.300	4.642	7.734	4.483	5.881	9.165
0.3	0.970	2.939	4.252	7.236	4.199	5.547	8.664	-0.3	0.970	3.230	4.569	7.649	4.449	5.831	9.072
0.3	0.960	2.878	4.173	7.154	4.143	5.479	8.569	-0.3	0.960	3.186	4.513	7.569	4.418	5.793	8.996
0.3	0.950	2.816	4.104	7.055	4.090	5.416	8.486	-0.3	0.950	3.144	4.477	7.508	4.387	5.760	8.944
0.3	0.900	2.615	3.876	6.798	3.883	5.149	8.230	-0.3	0.900	2.994	4.311	7.335	4.266	5.606	8.763
0.3	0.850	2.468	3.708	6.644	3.711	4.969	8.022	-0.3	0.850	2.885	4.181	7.215	4.163	5.482	8.616

continued on next page

		$\alpha_L = 0$			$\alpha_L < 0$					$\alpha_L = 0$			$\alpha_L < 0$		
β_L	β_U	10%	5%	1%	10%	5%	1%	β_L	β_U	10%	5%	1%	10%	5%	1%
0.3	0.800	2.351	3.582	6.491	3.581	4.820	7.843	-0.3	0.800	2.792	4.064	7.101	4.057	5.376	8.500
0.5	0.995	3.177	4.507	7.586	4.324	5.683	8.872	-0.5	0.995	3.564	4.918	8.140	4.602	6.016	9.352
0.5	0.990	3.056	4.385	7.389	4.261	5.604	8.744	-0.5	0.990	3.486	4.844	8.024	4.591	6.003	9.318
0.5	0.980	2.916	4.218	7.183	4.170	5.485	8.608	-0.5	0.980	3.403	4.748	7.892	4.565	5.965	9.273
0.5	0.970	2.817	4.100	7.057	4.084	5.390	8.463	-0.5	0.970	3.342	4.684	7.804	4.539	5.929	9.236
0.5	0.960	2.742	4.013	6.962	4.013	5.312	8.346	-0.5	0.960	3.298	4.641	7.746	4.516	5.893	9.190
0.5	0.950	2.677	3.942	6.869	3.946	5.227	8.261	-0.5	0.950	3.253	4.602	7.695	4.493	5.865	9.139
0.5	0.900	2.445	3.674	6.559	3.686	4.933	8.012	-0.5	0.900	3.120	4.444	7.519	4.392	5.737	8.947
0.5	0.850	2.276	3.484	6.343	3.494	4.729	7.735	-0.5	0.850	3.017	4.335	7.390	4.304	5.631	8.804
0.5	0.800	2.155	3.333	6.169	3.340	4.553	7.533	-0.5	0.800	2.932	4.242	7.266	4.212	5.541	8.679
0.7	0.995	3.056	4.377	7.431	4.245	5.618	8.798	-0.7	0.995	3.671	5.052	8.279	4.721	6.137	9.495
0.7	0.990	2.909	4.212	7.235	4.151	5.496	8.645	-0.7	0.990	3.603	4.968	8.182	4.710	6.128	9.467
0.7	0.980	2.742	4.015	6.971	4.015	5.319	8.398	-0.7	0.980	3.522	4.878	8.082	4.689	6.102	9.429
0.7	0.970	2.632	3.885	6.799	3.899	5.176	8.259	-0.7	0.970	3.467	4.827	8.012	4.666	6.068	9.383
0.7	0.960	2.534	3.773	6.689	3.791	5.057	8.131	-0.7	0.960	3.424	4.792	7.946	4.646	6.039	9.360
0.7	0.950	2.452	3.677	6.589	3.701	4.956	8.011	-0.7	0.950	3.388	4.755	7.878	4.630	6.020	9.334
0.7	0.900	2.191	3.375	6.205	3.381	4.595	7.637	-0.7	0.900	3.268	4.621	7.732	4.543	5.905	9.196
0.7	0.850	2.005	3.150	5.937	3.156	4.355	7.284	-0.7	0.850	3.176	4.506	7.614	4.465	5.819	9.044
0.7	0.800	1.855	2.962	5.708	2.975	4.155	7.036	-0.7	0.800	3.102	4.429	7.519	4.391	5.743	8.939